UNIVERSITY OF AMSTERDAM
Economics & Business

Income gap of immigrants in Pennsylvania

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Introduction

This empirical study aims at understanding the effects of economic variables on income in the state of Pennsylvania, with a focus on the citizenship status of individuals. The basic models investigated the effect of citizenship status, and the augmented models observed the effects of other variables.

Data

For data preparation, we followed the standard steps. The data was filtered to have entries from the state of Pennsylvania. Then we filtered for positive income and working-age population (18 to 64 years, inclusive).

We summarized the “educ” (education) variable by creating a new variable, "years of education". Years of education is a discrete numerical variable in the range [0,9], representing the number of years of education past middle school. For instance, value one represents grade 9, value four represents grade 12, and value 7 represents three years of college. The years of education is capped at 5 years of college.

Also, we decided to merge American-born citizens with those born to American parents abroad into one category variable since the latter category had few observations (0.66% of entries), and we suspected minimal change in study results.

Analysis

The analysis was initially performed using a simple OLS model with “Years of Education”, “age”, “sex”, “Citizenship Status”, and “Years in USA” regressors. We use robust regression with relaxed confidence intervals to account for heteroskedasticity of the error terms. In the Augmented Model, other variables were analysed as well.

Basic model

After running the regression, we have obtained the following results.

Table 1: Basic model VARIABLES		Income (logarithmic)
General Information	Years of Education	**0.187*** **
		(0.00214)
	Age	**0.00554*** **
		(0.000410)
	Is female	**-0.473*** **
		(0.00904)
Citizenship Status	Naturalized Citizen	**0.0407**
		(0.0337)
	Immigrant	**-0.223*** **
		(0.0345)
	Years in USA	**0.000636**
		(0.000952)
	Constant	**9.561*** **
		(0.0250)
	Observations	55,185
	R-squared	0.160

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

The general information features explain income through age, education and sex. We can observe that each extra year of education induces a 18.7% increase in total income, ceteris paribus. Each additional year of age increases income by around 0.554%, and females seem to earn around 46.6% less than males, on average. (Table 1)

For our citizenship status variables, we have US citizens (Those born in the United States or to American parents) as our benchmark, to avoid multicollinearity issues while one-hot-encoding the citizenship categories. The choice of benchmark follows the abundance of American citizens in the sample. Therefore, our analysis indicated that immigrants in the United States earn approximately 22.3% less, whereas being a naturalised citizen increases income by 0.4%, relative to being a US born. Thus, the gap is largest between naturalised Americans and non-

citizens. The results for naturalised citizens were not statistically significant, suggesting a minimal difference between U.S.-born and naturalized citizens.

The other coefficient that is not statistically significant (p-value > 0.1) is Year in the USA. As this variable is set to zero for US citizens and non-zero for other nationalities, this result follows the skewness of the variable.

Finally, our achieved R-squared for the initial model is 16 %. Although the explained variation is not high, the model itself is highly interpretable, and the results seem to follow experimenter predictions and economic theory. Thus, we conclude that the initial model provides a decent understanding of the relationship between people's incomes and different variables in the dataset for Pennsylvania.

Augmented model

We iteratively added features to the models and compared R2 scores and adjusted R2 scores. The first augmented model included the interaction of years in the USA with citizenship status, being an immigrant or a naturalized citizen. Unlike the basic model, in this construction, years in the USA have a significant and positive effect on income. (Table 2)

In the second augmented model, we made a dummy variable to control for urban-rural income differences, thresholding on the 80th percentile of population density. As a hypothesis, we think that incomes tend to be higher in densely populated areas. Moreover, we included employment status as a categorical variable. We witnessed the highest jump in R2 score with these additions, from 16% to 32%. Living in a big city increases income by approximately 19.4%, confirming the urban wage premium.

In the third model, we added racial dummy variables to account for potential ethnic income gaps. The baseline is white Americans, who make up 88 percent of the sample population. Even though most ethnic categories have significant coefficients, the R2 increases minimally. The highest effects were for Pacific islanders and African Americans, with 44% and 20% less income on average, respectively.

Table 2: Augmented Models VARIABLES		(Basic) Log income	(Aug1) Log Income	(Aug2) Log Income	(Aug3) Log Income
General Information	Years of Education	**0.187*** **	**0.188*** **	**0.147*** **	**0.144*** **
		(0.00214)	(0.00214)	(0.00197)	(0.00197)
	Age	**0.00554*** **	**0.00534*** **	**0.0121*** **	**0.0117*** **
		(0.000410)	(0.000411)	(0.000372)	(0.000372)
	Is female	**-0.473*** **	**-0.473*** **	**-0.430*** **	**-0.429*** **
		(0.00904)	(0.00904)	(0.00813)	(0.00812)
Citizenship Status	Naturalized Citizen	**0.0407**	**-0.0731**	**-0.145*** **	**-0.124** **
Baseline: US Citizen		(0.0337)	(0.0547)	(0.0504)	(0.0521)
	Immigrant	**-0.223*** **	**-0.389*** **	**-0.336*** **	**-0.318*** **
		(0.0345)	(0.0507)	(0.0457)	(0.0475)
	Years in USA	**0.000636**			
		(0.000952)			
Interaction of Years in the USA with Citizenship	Naturalized Years in the USA		**0.00501*** **	**0.00574*** **	**0.00532*** **
			(0.00191)	(0.00174)	(0.00174)
	Immigrant Years in the USA		**0.0143*** **	**0.00843*** **	**0.00865*** **
			(0.00313)	(0.00280)	(0.00280)
Employment Status	Is from a Dense Area			**0.188*** **	**0.190*** **
Baseline: Unemployed				(0.0327)	(0.0325)
	Employed			**1.367*** **	**1.356*** **
				(0.0172)	(0.0173)
	Has a job, not working			**0.977*** **	**0.971*** **
				(0.0451)	(0.0452)
	Armed Forces			**1.624*** **	**1.611*** **
				(0.0820)	(0.0819)
	Unemployed			**0.185*** **	**0.189*** **
				(0.0495)	(0.0494)
Race	Asian				**0.0407**
Baseline: white					(0.0274)
	Native American				**-0.0478**
					(0.0326)
	African American				**-0.202*** **
					(0.0183)
	Pacific Islander				**-0.443* **
					(0.238)
	Other				**-0.0824*** **
					(0.0205)
	Constant	**9.561*** **	**9.568*** **	**9.690*** **	**9.733*** **
		(0.0250)	(0.0250)	(0.0224)	(0.0225)
	Observations	55,185	55,185	55,185	55,185
	R-squared	0.1597	0.1602	0.3250	0.3273
	Adjusted R-Squared	0.1596	0.1601	0.3248	0.3270

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Conclusion

Regarding our central research question, we observe that citizenship status has an important effect on income. Immigrants earn less than naturalized and U.S.-born citizens. Being a naturalized citizen had a statistically insignificant effect on income, as opposed to being U.S.-born. Other augmented models did better in explaining the data than the restricted basic model and followed economic theory and experiment predictions. Years spent in the United States variable was insignificant on its own, but its interaction with citizenship status provided additional information and became significant. This is highly likely due to the fact that the years in the USA variable was set to 0 for U.S.-born individuals.

References

- Ruggles, Steven et al. (2025). “IPUMS USA: Version 16.0”. In: [dataset]. doi: 10.18128/D010.V16.0.url: https://doi.org/10.18128/D010.V16.0.

Appendix

- Git hub page for the Stata do-file and additional information: https://github.com/DovydasVe/Econometric_analysis_income_gaps
- DO file:

```
* Load the dataset
use "assignment_data.dta"
set seed 42

* Filter for Pennsylvania
keep if stateicp==14

* Filter for appropriate age groups
drop if age>64
drop if age<18

* Positive total income
drop if inctot<0

* Check the categories in education, recode them for degree
label list EDUC
recode educ (0/2 = 0) (3/5 = 1) (6 = 2) (7/8 = 3) (9/11 = 4), gen(degree)

* Another variable explaining the education, is years of education. Years of
education is discrete numerical variable in range [0,9], representing the
number of years of education past middle school.
* Value one represents grade 9, value four represents grade 12, value 7
represents three years of college.
. gen yreduc = educ - 2
. replace yreduc = 0 if educ<3



* Generate dummy variables for citizen categories, merge those born abroad
* with American citizens. We choose to use manual one-hot-encoding over stata
builtin methods for ease of interpretation.


gen cit_1 = 1 if citizen == 0 | citizen == 1
replace cit_1=0 if citizen == 2 | citizen == 3
gen cit_2 = 1 if citizen ==2
```

```
replace cit_2=0 if citizen != 2
gen cit_3 = 1 if citizen ==3
replace cit_3=0 if citizen != 3

* Cit_1 refers to American Citizens, either born in US or to American parents
* Cit_2 refers to naturalized citizens in the US
* Cit_3 refers to not american citizens
* Take the log of income as a response variable
gen log_inctot = ln(inctot)

* Regress log of income on degree, age, sex, and citizenship status. Cit_1
refers to
reg log_inctot yreduc c.age i.sex cit_2 cit_3 yrsusa1, robust

* Improving the model with distinction of where people live (big city or
not) depending on density
sort density
gen cumprob = sum(density)
replace cumprob = cumprob / cumprob[_N]
local cutoff = (0.7926005 + 0.7928035) / 2
gen big_city = (cumprob > `cutoff')

* Racial distinctions, making them binary var. (0,1)
gen asian = racasian - 1
gen amind = racamind - 1
gen black = racblk - 1
gen pacis = racpacis - 1
gen white = racwht - 1
gen other = racother - 1

* The final model will include big_city, racial differences and i.empstatd
* The greatest improvement on adjusted R squared was due to adding empstatd,
which is logical
* As the distinction between working people and not working people can help
in predicting one's income
* Additionally, we added degfield to differentiate people based on degree or
field they work in


 gen yrsusa_cit2 = yrsusa1 * cit_2
 gen yrsusa_cit3 = yrsusa1 * cit_3

 regress log_inctot yreduc c.age i.sex cit_2 cit_3 yrsusa_cit2 yrsusa_cit3,
robust

 reg log_inctot yreduc c.age i.sex  cit_2 cit_3  yrsusa_cit2 yrsusa_cit3
big_city i.empstatd , robust

reg log_inctot yreduc c.age i.sex  cit_2 cit_3  yrsusa_cit2 yrsusa_cit3
big_city i.empstatd asian amind black pacis other , robust
```